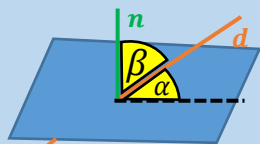


Angles between Lines and Planes

Key Points

α - angle between line and plane



For a line with direction vector, d and a plane with normal, n . Use

$$d \cdot n = |d||n| \cos \beta$$

Then use $\alpha = 90 - \beta$, or instead use

$$d \cdot n = |d||n| \sin \alpha$$

Be careful for differently oriented normal and directions

Basic Skills

Q1. For $a = (1,2,3)$, $b = (0,5,2)$, $c = (-1,3,-5)$, calculate the following

a) $a \cdot b$ b) $a \cdot c$ c) $c \times b$ d) $|c|$

e) The angle between a , b

f) The normal to the two vectors, a , b

Q2. $r = p + \lambda(1,2,3)$ and $r = (\lambda - 2, 2\lambda, 5 + 3\lambda)$ both describe the same line, find the constant vector p

Q3. Work out the angle between $a = (4,3,0)$ and $b = (1,3,-5)$

Q4. Find the angle between the lines $r = p + \lambda(1,2,3)$ and $r = p + \lambda(2,-4,1)$ for some unknown p

Competency Skills

Q5. For the line with direction vector $d = (2,4,1)$ and a plane with a normal $n = (5,7,10)$

a) Find $d \cdot n$, $|d|$ and $|n|$

b) Find the angle between the direction and normal and hence the angle between the line and plane

Q6. For the line $r = \begin{pmatrix} 5 \\ 1 \\ -1 \end{pmatrix} + t \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$ and the plane with points $(5,4,2)$, $(-2,6,10)$ and $(0,1,4)$

a) Work out the normal to the plane

b) Hence work out the angle between the line and plane

Q7. Find the angle between $r = \begin{pmatrix} 4 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ and the plane $6x - y + 3z = 80$

Advanced Skills

Q8. Find the angle between the line $r = (2 + t, 3 - 2t, 4t)$ and the plane $r = a + \lambda(1,0,1) + \mu(1,-2,1)$ for some unknown constant vector a

Q9. If the line is in-planar what is the angle between the line and plane and hence what is the scalar product

Q10. A student suggests that "There are infinite angles between any two vectors as $\cos \theta = c$ has infinite solutions, when no interval is given". Amend their statement to be correct

Q11. Find the angle between the two planes $5x - y = 11$ and $6x - 8y + z = 0$

Extension

Q13. Explain why the normal to the plane is used instead of in plane directions when working with planes and angles